Performance practice exercises

E1 Lighthouse | E2 responsive images | E3 script loading

This report explains the three working examples, the code decisions and the checks used to verify them. Open the exercise collection.

E1 / Run Lighthouse and read the results

The starter scored **77/100** for Performance; the optimized page scored **100/100**. The longest time-based starter metric was Largest Contentful Paint (LCP), at **5.0 s**. CLS is unitless, so it is not compared numerically with milliseconds or seconds.

Metric	Before	After
First Contentful Paint	1.5 s	0.6 s
Largest Contentful Paint	5.0 s	1.2 s
Speed Index	2.4 s	0.6 s
Total Blocking Time	0 ms	0 ms
Cumulative Layout Shift	0.116	0.001

Changes made

Removed the unused blocking jQuery dependency; used a small external deferred app script; served smaller local images; added responsive candidates and image dimensions; and lazy-loaded the lower gallery while keeping the primary image eager.

```
<script src="e1-app.js" defer></script>
<img src="../assets/photo-800.webp" width="800" height="400"
     alt="Two yellow trams on tracks along a Lisbon street"
     fetchpriority="high">
<img src="../assets/movie3.jpg" width="400" height="560"
     alt="A winding path along a wooded coastline" loading="lazy">
```

Lighthouse 13.6.0; mobile emulation; simulated throttling (150 ms RTT, 1,638.4 Kbps, 4x CPU slowdown); one navigation run per version on the same local server. The optimized exercise replaces random remote photographs with local sample assets; network and redirect differences contribute. This comparison does not isolate the effect of defer.

E1 findings and code | Optimized practice page

E2 / Responsive images

WebP source, JPEG fallback and two usable image widths.

The picture element offers WebP first and a JPEG fallback. Both use 400w and 800w candidates. The browser combines sizes, viewport width and pixel density when choosing a file. Explicit width and height reserve a 2:1 aspect ratio.

```
<picture>
  <source type="image/webp"
    srcset="../assets/photo-400.webp 400w,
            ../assets/photo-800.webp 800w"
    sizes="(max-width:650px) calc(100vw - 36px), 800px">
  <img src="../assets/photo-800.jpg"
    srcset="../assets/photo-400.jpg 400w,
            ../assets/photo-800.jpg 800w"
    sizes="(max-width:650px) calc(100vw - 36px), 800px"
    alt="Two yellow trams on tracks along a Lisbon street"
    width="800" height="400" fetchpriority="high">
</picture>
```

Measured file sizes

Dimensions	JPEG	WebP	Reduction
400 x 200	21,448 bytes	13,466 bytes	37.2%
800 x 400	53,204 bytes	28,346 bytes	46.7%

These are the actual supplied files, encoded at JPEG quality 85 and WebP quality 80. Savings depend on image content and encoding settings.

What was verified

At a 390px viewport, the browser selected photo-400.webp at device pixel ratio 1 and photo-800.webp at pixel ratio 2. The page displays currentSrc so the choice can be inspected. A separate plain JPEG image verifies fallback loading.

Open E2 and inspect the selected image

E3 / defer and async

Use external files so the loading attributes have an effect.

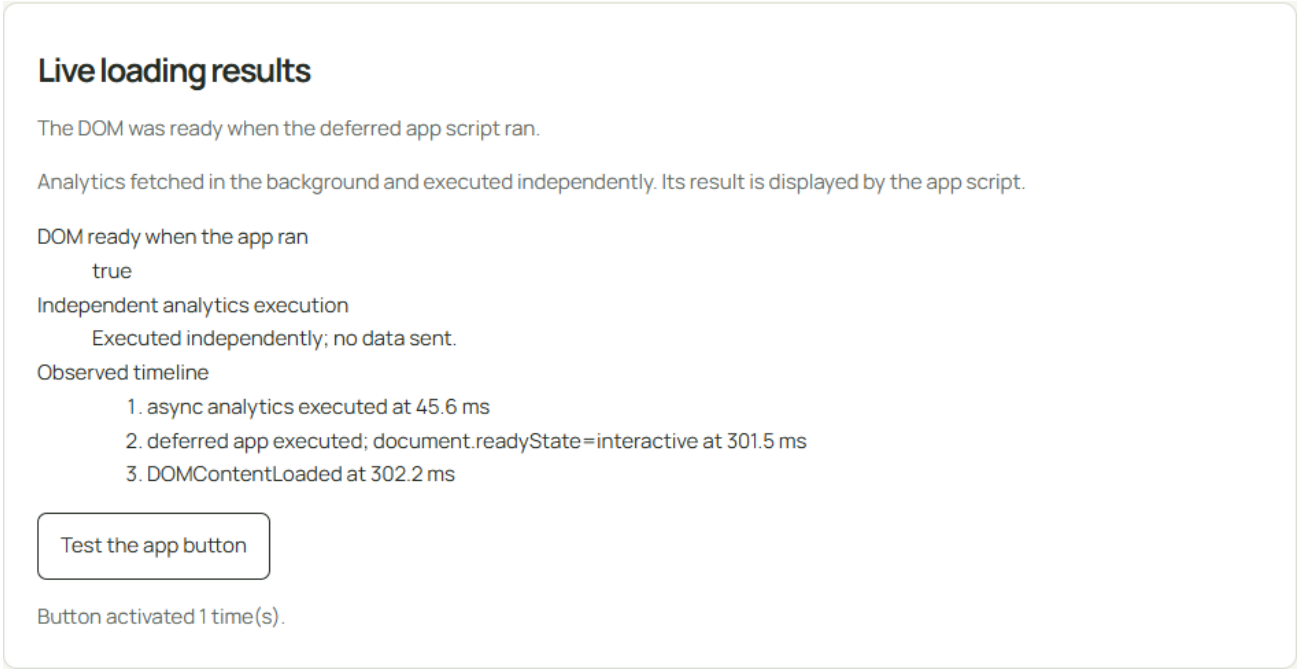
The app uses the DOM and is loaded with defer. The independent analytics example uses async; it records an execution event without reading the DOM or sending data. Both script tags are in head.

```
<head>
  <script src="app-script.js" defer></script>
  <script src="analytics.js" async></script>
</head>
```

Attribute	What it does
defer	Downloads alongside parsing; executes after parsing and preserves document order.
async	Downloads alongside parsing; executes when ready, with no guaranteed relative order.

The original exercise used inline classic scripts. defer and async do not defer those inline scripts, so the app and analytics code were extracted into separate files. The simulated busy-wait was removed.

Observed browser proof

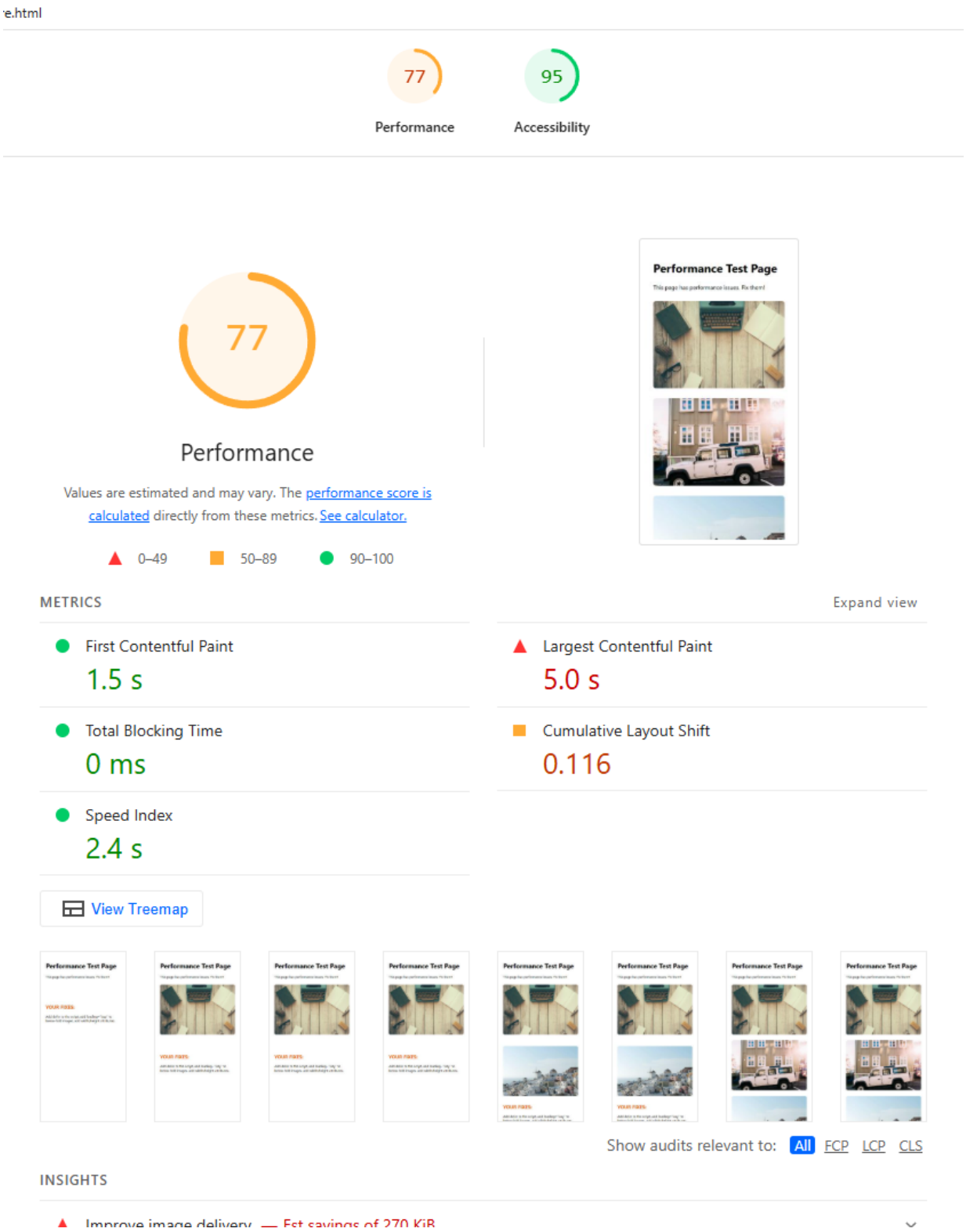


This screenshot records one run. The async timing and relative order may change between runs. The app status confirms the DOM exists, and the button responds after loading.

Open E3 and rerun the timeline

E1 / Before: 77 out of 100

Original saved Lighthouse report: score, metrics and page preview.

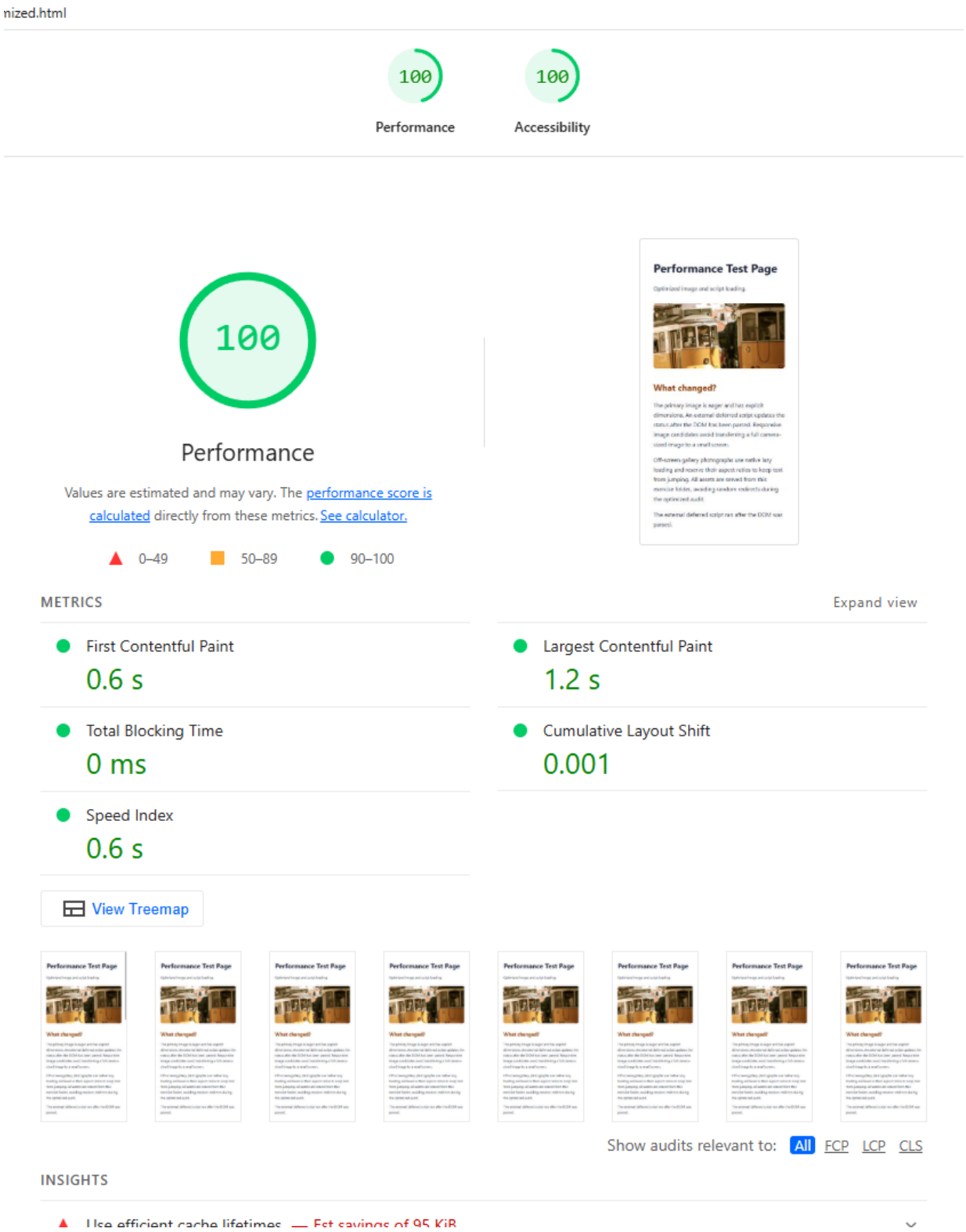


Audit URL: http://127.0.0.1:4173/week9_2/exercises/E1_Before.html
Captured: 2026-11-02T14:36:15.280Z (UTC).

Open the full HTML report | Open raw audit JSON

E1 / After: 100 out of 100

Original saved Lighthouse report: score, metrics and page preview.



Audit URL: http://127.0.0.1:4173/week9_2/exercises/E1_Optimized.html
Captured: 2026-11-02T14:37:07.229Z (UTC).

Open the full HTML report | Open raw audit JSON